



# Pip Architecture — F5: the agent loop (plan)

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**Status:** plan → build, on branch `claude/f5-agent-loop-gtbcok` . NOT deployed; Chris fast-forwards `main` after he tests. This is the last open item in the X6 FORM sequence (F1 shared builder → F2 persist → F3 event-driven → **F5 agent loop** → F6 pgvector recall).

**One-line goal:** give Pip's chat a real tool-use loop — model emits a tool call → server executes it → returns the result → model keeps reasoning → final answer — so Pip can do multi-step work in one turn ("find the stalled project, then draft the chase note") instead of one isolated action.

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## The #1 rule: do not regress today's chat path

Today's chat is **single-shot**: the model emits text + `tool_use` blocks, the server streams the text and forwards the `tool_use` blocks to the client, and the **client** executes them (frictionless ones fire immediately; the rest go through a confirm card). The tool *result* never goes back to the model. ( `api/pip.js` streaming branch → `src/lib/pipStream.js` → `PipView.send` → `planToolCalls / executeTool` .)

That single-shot behavior is the **floor**. The loop is additive and degrades to it cleanly:

- The loop activates **only** when the model emits a *read tool* (a new, server-executed, read-only tool family). If the model emits no read tool — which is every chat turn today — the loop runs exactly one model call and behaves byte-identically to today.
- A kill switch ( `PIP_AGENT_LOOP=off` ) removes read tools from the tools array entirely, so the model *can't* call them → guaranteed single-shot.

- On any loop error or cap, we land on a clean single text answer — never a broken turn.
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## Why the loop needs NEW read tools (the key architectural call)

Pip's *existing* tools ( `src/lib/pipTools.js` ) cannot be the thing the loop iterates on, for two reasons:

1. **They execute client-side.** `open_*` / `navigate` are pure UI actions with no meaningful "result." The DB-write tools ( `create_open_item` , `log_meeting` , ...) run against the user's Supabase session via React hooks in `PipView` . The server has no way to produce a `tool_result` for them.
2. **Executing writes server-side would regress the confirm-card safety model.** Today a write goes through a confirm card ( `needsConfirm` → `category: "confirm"` ). Running it server-side mid-loop would silently commit it. Unacceptable.

So the existing **action tools stay exactly as today: client-executed and *terminal* within a chat turn.** The loop iterates on a small, new family of **read-only, server-executed tools** that let Pip *gather* mid-turn. A turn is naturally either "gathering" (read tools → loop continues) or "acting" (an action tool → terminate, hand to client) — matching how Pip should work: look first, then act.

This is the honest reading of "expose existing tools": the existing tools' behavior is preserved verbatim; we add the minimal read surface a real loop requires.

### The read tools (v1)

All three are read-only, scoped to the caller via the JWT-attached Supabase client (RLS does the rest), and return the user's **own notebook data** — which the Data Line Rule explicitly permits Pip to read (the rule governs Pip *soliciting* and *retaining* business data, not reading Chris's own verbatim notes). None solicit or persist numbers.

| Tool                        | Input                                                                                          | What it returns                                                                                                                                                                | Renderer                                                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>lookup_account</code> | <code>account_name</code>                                                                      | Deep current detail for ONE account: recent meetings (with notes), open items, contacts, active projects. For accounts not in the trimmed context, or deeper than the summary. | <code>accountContext.renderAccount({surface:"chat"})</code> — the F1 should PARITY by construction. |
| <code>find_open_work</code> | <code>filter ∈<br/>overdue  <br/>stalled  <br/>waiting_on_them<br/>  due_soon  <br/>all</code> | Concise cross-portfolio rows (account · what · status · age · who-has-ball). Serves "find the stalled project / what's slipping / who has the ball."                           | compact line list (query result, no context — no parity surface)                                    |
| <code>search_notes</code>   | <code>query</code>                                                                             | Matching snippets from the user's meeting                                                                                                                                      | compact snippet list                                                                                |

|  |                                                                                                                                          |
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|  | notes + Pip<br>summaries<br>across all<br>accounts<br>(account ·<br>date ·<br>snippet).<br>Serves<br>"where/when<br>was X<br>discussed." |
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Read tools **never cross to the client** — they're a server-internal working set. The client only ever receives action tools (it has no executor for read tools). This is a hard invariant in the forwarding code.

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## WHERE the loop lives

- **src/lib/pipAgentTools.js** (NEW, pure — no Supabase/React/fetch, unit-tested): the read-tool definitions ( `PIP_READ_TOOLS` ), `isReadTool` , `partitionToolUses(content) → {readTools, actionTools, text}` , and `decideLoopStep({partition, step, maxSteps}) → "continue" | "terminate" | "force_final"` . The risky control logic is pure and tested here.
- **api/\_pipAgentLoop.js** (NEW, server I/O — receives the Anthropic `client` as a param, so it does NOT import the SDK and stays clear of Guard 3): `executeReadTool(name, input, ctx)` (Supabase queries + `accountContext` render) and `runAgentChat({...})` (drives the loop using the pure helpers).
- **api/pip.js** (chat streaming + buffered branches): when chat mode + loop enabled, route through `runAgentChat` ; otherwise the existing inline single call, untouched. `tools` becomes `PIP_TOOLS.concat(PIP_READ_TOOLS)` only when the loop is on.

`.js` extensions on all relative imports (CLAUDE.md API Module Import Rule).

`api/_pipAgentLoop.js` is a helper (like `_pipUsage.js`), not a handler, so it needs no entry in `test-api-imports.js`, but it's import-tested transitively when `api/pip.js` loads.

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## The loop (mechanics)

```
tools = PIP_TOOLS + PIP_READ_TOOLS          // byte-stable across all
requests & iterations
system = [PIP_STATIC_SYSTEM (cache_control), dynamic tail]  //
identical object reused every iteration
messages = [ ...chat history ]

for step in 0 .. maxSteps-1:
  forceFinal = (step === maxSteps-1)
  params = { model, max_tokens, system, tools, messages,
            tool_choice: forceFinal ? {type:"none"} : undefined }
// none keeps tools+system cache valid
  stream = client.messages.stream(params)
  stream.on("text", delta => sseDelta(delta))          // stream EVERY
turn's text live
  final   = await stream.finalMessage()
  logPipUsage(... final.usage ...)                    // log EACH model
call → spend tile sees full cost

  { readTools, actionTools } = partitionToolUses(final.content)

  // Continue ONLY when the turn is pure gathering: read tools
  present AND no action tools.
  if (readTools.length === 0 || actionTools.length > 0 ||
forceFinal):
    return { fullText, actionToolCalls: actionTools }  //
TERMINAL

  messages.push({ role:"assistant", content: final.content })
// includes the read tool_use blocks
  results = await Promise.all(readTools.map(executeReadTool))
// each → tool_result block (is_error on failure)
  messages.push({ role:"user", content: results })
// every tool_use answered → API satisfied
```

Then the handler emits the existing SSE shape: `delta` events already streamed, one `tool_use` event per **action** tool (read tools filtered out), and `done` with the accumulated `content` + action `tool_calls`. **Zero client changes** — `pipStream.js` and `PipView` see the same protocol they see today.

### Why "terminate when any action tool appears"

The API requires a `tool_result` for every `tool_use` in the prior assistant turn. If a turn mixed a read tool and an action tool, continuing the loop would leave the action tool's `tool_use` unanswered (we can't run it server-side). So a turn with any action tool is **terminal** — all action tools go to the client, and any read tool in that same terminal turn is simply not executed (rare; the model already has enough to act). The loop continues only on pure read-only turns. Clean, and no dangling `tool_use`.

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## HARD cost guard

- **Step cap:** `maxSteps` = 4 model calls ( $\leq 3$  read round-trips), env-overridable via `PIP_AGENT_MAX_STEPS`. On the final allowed step, `tool_choice: {type: "none"}` forces a text answer — graceful landing, never a half-finished loop.
- **Per-call ceiling:** `max_tokens` stays 900 (chat config) per call.
- **Full visibility:** `logPipUsage` fires on **every** model call, so the Settings spend tile and `folio_pip_usage` reflect true loop cost (no invisible spend — the failure class Batch 1 fixed).

**Worst-case cost per chat turn** (Sonnet \$3/M in, \$15/M out; chat `max_tokens` 900):

- A turn that *doesn't* loop (the overwhelming majority — same as today): **1 call**.
- Absolute worst case: 4 calls. Per call  $\approx$  static system (1.4k tok, ~~cache-read at 0.1x after call 1~~) + context tail + growing `tool_result` blocks (uncached, est. 4-8k tok) +  $\leq 900$  output.  $\approx (6k \times 0.3 + 900 \times 1.5) / 1000 \approx \mathbf{\$0.0045/M-}$

**scaled  $\approx$  ~\$0.003/call  $\rightarrow$  ~\$0.012 worst case for a fully-looped turn.**

Even pessimistically padding context to 12k input/call, worst case stays **under ~\$0.03/turn**, and only on turns where Pip genuinely needed 3 lookups. The step cap makes runaway impossible; the daily spend cap ( `overDailySpendCap` ) remains the portfolio-level backstop.

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## Streaming

Chat streams today. The loop preserves it: **text deltas stream live on every iteration** (the brief "let me check that project..." preamble of a gathering turn, then the final answer), accumulating into `done.content` . Tool round-trips happen entirely server-side — the client never sees a read tool or a `tool_result` . The SSE protocol ( `meta / delta / tool_use / done / error` ) is unchanged, so `pipStream.js` needs no edits.

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## Prompt cache discipline

- `tools` (= `PIP_TOOLS` + `PIP_READ_TOOLS` ) is byte-stable across all requests and every loop iteration  $\rightarrow$  the tools cache prefix holds.
  - The same `system` blocks object (static block carries `cache_control:ephemeral` ) is reused on every iteration — never rebuilt mid-loop  $\rightarrow$  system cache holds from iteration 2 onward.
  - Only `messages` grows (appended assistant `tool_use` + user `tool_result` ); the cached `tools` + `system` prefix is read, not rewritten. `tool_choice:none` on the forced-final call invalidates only the messages tier, not tools/system (per the caching invalidation hierarchy).
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## Fallback (current behavior is the guaranteed floor)

- **Loop disabled** ( `PIP_AGENT_LOOP=off` ): read tools are not added to the tools array; the model can't call them; the existing single inline call runs → byte-identical to today.
  - **Read-tool execution error:** returns a `tool_result` with `is_error:true` + a short message → the model recovers or answers without it. Never throws the turn.
  - **Cap reached:** the final step runs with `tool_choice:none` → a clean text answer.
  - **Stream/model error mid-loop:** caught; if text was already produced, emit `done` with it; else emit the existing `error` event. Same as today's catch.
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## Parity (F1)

`lookup_account` renders the deep account read through `accountContext.renderAccountContext(bundle, {surface:"chat"})` — the one shared builder F1 introduced — so a field added to that builder reaches the loop for free, exactly as it reaches chat/brief/summarize/operator. No new bespoke account renderer.

`find_open_work` / `search_notes` return query rows, not "what Pip knows about an account," so they don't touch the parity surface.

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## Data Line Rule

Read tools read Chris's own verbatim notebook — explicitly allowed. They **never solicit** numbers (tool descriptions ask for account names / topics / status filters, never

figures) and **never retain** anything (reads write nothing). Documented in `ai-governance.md` (how the agent loop is bounded) + `data-handling.md`.

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Tests (drift + behavior locks; node-only vitest env, so logic is pure-tested)

New `src/lib/pipAgentTools.test.js` :

- `PIP_READ_TOOLS` shape (name/description/input\_schema present; all read-only).
- `isReadTool` correctly classifies read vs action tool names.
- `partitionToolUses` splits a mixed content array into read/action/text.
- `decideLoopStep` : pure read-only turn → `continue` ; turn with an action tool → `terminate` ; no tools → `terminate` ; last step → `force_final` .
- Invariant: an action-tool-only turn never continues (no-regression lock).

Existing 330 tests must stay green. Gates every commit: `vite build` · `vitest` · `node scripts/check-guards.js` · `node scripts/test-api-imports.js` .

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Out of scope (deliberately)

- Server-executing write/action tools (would regress the confirm card).
- Client changes to `pipStream.js` / `PipView` (the loop is server-internal).
- F6 pgvector recall (separate item; `search_notes` here is plain ilike, not semantic).
- Looping the non-chat modes (summary/brief/email/action) — those keep today's behavior exactly.